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Apparatus, system, and method of providing a dispenser of circuit board component underfill

Abstract

An apparatus, system and method for dispensing underfill to components on a printed circuit board. The apparatus, system, and method includes a dispensing end effector suitable for dispensing underfill to components on a printed circuit board within an underfill chamber. The dispensing end effector may include: an electromechanical connection to at least one dispensing robot arm capable of physically situating the dispensing proximate to the circuit board; a communicative connection to a dispense controller capable of communicatively controlling at least the dispensing; a dispenser which includes an underfill output port, which is capable of the dispensing, and which is removably mounted to the electromechanical association; and a protective enclosure at least substantially about the dispenser.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims benefit of priority to U.S. Provisional Application No. 62/788,511, filed Jan. 4, 2019, entitled: “Apparatus, System and Method of Providing a Dispenser of Circuit Board Component Underfill,” the entirety of which is incorporated herein by reference as if set forth in its entirety.

BACKGROUND

Field of the Disclosure

(1) The disclosure relates generally to circuit board manufacturing, and, more particularly, to an apparatus, system, and method of providing underfill on a circuit board.

Background of the Disclosure

(2) In a typical printed circuit board production process, numerous steps must be performed in order to complete the subject board. In short, these steps typically include the placement of solder on the board adjacent and along the existing printed circuit traces; the picking and placement of circuit components onto the solder, wherein such components may include, by way of example, resistors, capacitors, inductors, transistors, diodes, integrated circuit chips, and the like; reflow such that the placed board components are secured to and electrically associated with the printed circuit traces; and the underfill of the placed components to provide mechanical support to the components.

(3) However, the underfill process suffers from enhanced difficulty for larger boards, and least because it becomes difficult to access the components on a large board to underfill those components, and further because, as underfill is the result, in most processes, of a capillary action by which the underfill flows under the component in the presence of heat. There are no known processes by which such heat can be consistently applied to large boards, particularly for large boards of odd shapes, prior to crosslinking by the underfill. There are also no known methodologies by which heat contamination within an underfill chamber can be prevented from causing crosslinking of the underfill while still within the underfill dispenser. Upon crosslinking by the underfill, the underfill will no longer move into the empty space beneath the components via the referenced capillary action, and if the crosslinking occurs while the underfill is still within the dispenser, clogging occurs such that the underfill cannot be properly dispensed.

SUMMARY OF THE DISCLOSURE

(4) The disclosure is and includes at least an apparatus, system and method for dispensing underfill to components on a printed circuit board. The apparatus, system, and method includes a dispensing end effector suitable for dispensing underfill to components on a printed circuit board within an underfill chamber. The dispensing end effector may include: an electromechanical connection to at least one dispensing robot arm capable of physically situating the dispensing proximate to the circuit board; a communicative connection to a dispense controller capable of communicatively controlling at least the dispensing; a dispenser which includes an underfill output port, which is

capable of the dispensing, and which is removably mounted to the electromechanical association; and a protective enclosure at least substantially about the dispenser.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure is illustrated by way of example and not limitation in the accompanying drawings, in which like references may indicate similar elements, and in which:

- (2) FIG. 1 illustrates a circuit board component;
- (3) FIG. 2 illustrates capillary action for an underfilled component;
- (4) FIG. 3 illustrates and underfill machine;
- (5) FIG. 4 illustrates and underfill machine;
- (6) FIG. 5 illustrates and underfill machine;
- (7) FIG. 6 illustrates an underfill dispenser;
- (8) FIG. 7 illustrate underfill dispensers;
- (9) FIG. 8 illustrate underfill dispenser;
- (10) FIG. 9 illustrates an underfill calibration system; and
- (11) FIG. 10 illustrates an underfill processing system.

DETAILED DESCRIPTION

(12) The figures and descriptions provided herein may have been simplified to illustrate aspects that are relevant for a clear understanding of the herein described devices, systems, and methods, while eliminating, for the purpose of clarity, other aspects that may be found in typical similar devices, systems, and methods. Those of ordinary skill may recognize that other elements and/or operations may be desirable and/or necessary to implement the devices, systems, and methods described herein. But because such elements and operations are well known in the art, and because they do not facilitate a better understanding of the present disclosure, a discussion of such elements and operations may not be provided herein. However, the present disclosure is deemed to inherently include all such elements, variations, and modifications to the described aspects that would be known to those of ordinary skill in the art.

(13) The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. For example, as used herein, the singular forms “a”, “an” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

(14) When an element or layer is referred to as being “on”, “engaged to”, “connected to” or “coupled to” another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to”, “directly connected to” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(15) Although the terms first, second, third, etc., may be used herein to describe various elements,

components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. That is, terms such as “first,” “second,” and other numerical terms, when used herein, do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the exemplary embodiments.

(16) Processor-implemented modules, systems and methods of use are disclosed herein that may provide access to and transformation of a plurality of types of digital content, including but not limited to video, image, text, audio, metadata, algorithms, interactive and document content, and which track, deliver, manipulate, transform and report the accessed content. Described embodiments of these modules, systems and methods are intended to be exemplary and not limiting. As such, it is contemplated that the herein described systems and methods may be adapted and may be extended to provide enhancements and/or additions to the exemplary modules, systems and methods described. The disclosure is thus intended to include all such extensions.

(17) FIG. 1 illustrates the top view of a component **102** placed onto a printed circuit board **104**. The component has, under-filled beneath the component and at least partially extending outside the outermost perimeter **106** of the component from a top view, an under-fill **110**. The under-fill **110** is generally placed, in part, because solder joints **112** formed by the reflowed solder to hold the component **102** to the board **104** do not provide sufficient mechanical strength to hold the component **102** sturdily and operably in place on the board **104**. The under-fill **110** beneath the component **102** and adjacent to the solder joints **112** provides this mechanical stability.

(18) FIG. 2 illustrates a capillary action **202** of an under-fill **110** such that the underfill **110** flows to support a board component **102**. In the illustration, about the perimeter **106** of the component **102** and adjacent to the reflow solder joints **112**, the under-fill **110** fills in by capillary action **202**. Application of heat **204** causes this creeping flow of the under-fill **110** by the capillary action **202** in the example shown. This creeping flow fills underneath the component **102** and into the gaps beneath and around the component **102**, between the component **102** and the board **104** on which it resides, and around the connectors by which the component **102** communicates with the board's traces through the solder joints **112**. Of note, it is typical that several applications of the under-fill **110** may be needed in order to sufficiently fill the under-component gaps such that the requisite mechanical stability is provided to the component **102** in its location on the subject board **104**.

(19) The various aforementioned aspects of under-fill **110**, namely, the providing of under-fill material upon the application of heat **204** that enables the capillary action **202** of the underfill **110**, are particularly difficult for larger size boards, as referenced above. This is, in part, because prior art under-fill machines have needed to be significantly bigger than the board placed therein, in order to allow for the referenced electromechanical components of the underfill machine to perform the various under-fill and heating functions described herein. Therefore, the larger board, the more the aforementioned issues of providing heat and underfill are exacerbated.

(20) Thus, the embodiments provide an apparatus, system, and method for an under-fill machine that accommodates circular boards of up to 48 inches or more in diameter to provide underfill to the components thereon. Accordingly, accommodated boards in the embodiments may also be smaller than 48 inches, such as 34 inches, 38 inches, or 42 inches, and the disclosed teachings may also be applied to boards larger than 48 inches, such as 52 inches, 54 inches, or 56 inches, by way of nonlimiting example.

(21) FIG. 3 illustrates an under-fill machine **302**. In the illustration, electromechanical/robotic elements **304** for the providing of at least underfill may be present within an underfill chamber **306**, such as two such robots to the right and left sides of the front aspect of the machine **302**. These robots **304** may be controlled by software **1490** executed by at least one processing system **1312**,

and this software **1490** may comprise algorithms that allow for training of the robots **304** to provide underfill in various contexts and to various components; that allow for the providing of this underfill; and for collision avoidance, such as in embodiments wherein the board **310** is sufficiently large so as to require the use of multiple robots **304** to provide underfill in multiple areas of the board **310** simultaneously, so that the robots **304** do not collide when transmitting their respective algorithmic paths.

(22) Also included, such as at the front aspect of the machine **302**, may be an accommodating input **314** of suitable size to allow for insertion into the machine **302** of a printed circuit board **310**, such as the larger boards discussed throughout. Of note, the accommodating input **314** may allow for placement of the subject board **310** onto a carrier **316** prior to and/or upon insertion to the input **314**, such that a printed circuit board **310** may or may not reside on the carrier **316** prior to and/or throughout the under-fill process, as discussed further herein. Additionally, the accommodating input **314** may receive the board **310** and/or the carrier **316** on which the board **310** resides via any known methodologies, such as manually or automatically, such as indexed or unindexed, and/or through the use of an input slide, shuttle or an input conveyor, by way of nonlimiting example.

(23) Now also with respect to FIGS. **3** and **4**, the chamber **306** of the under-fill machine **302** may include, proximate to the robots **304** and by way of nonlimiting example, one or more heat sources to heat the board, such as an overhead heat source **320** and a lower heat source **322**. The lower heat source **322** may include, by way of nonlimiting example, heating one or more aspects of the board **310**, or the entirety of the board **310**, from the underside of the board **310**, such as by heating, or providing heat from, the carrier **316** with which the board **310** is associated. This lower heat may be provided to the board **310** and/or to or from the carrier **316** via any known methodology, such as forced air heating, IR heating, or RF/inductive heating, by way of non-limiting example. It may be desirable in some embodiments that the lower heat provided be at least substantially uniform over that portion of the board **310** to which the heat is provided.

(24) The overhead heater **320** may be in the style of an oven, such as wherein the provided overhead translates through a medium, such as air or other gas, to the topmost portion of any aspect of the board **310** that is adjacent the overhead heater **320**. As such, the provided overhead heat may be of any known type, such as forced air heat, RF/inductively generated heat, and so on. Of note, in embodiments, the overhead heater **320** may only be of sufficient size to accommodate approximately half the circuit board **310** thereunder, such that the circuit board **310**, whether or not associated with the carrier **316**, may be rotated to place aspects previously unheated underneath the overhead heater **320**, such as one portion of the board **310** at a time. This rotation to be subjected to the overhead heater **320** may be performed manually or automatically, as discussed throughout, and is referenced herein throughout as underfill heat indexing.

(25) The foregoing indexing and heating may be subjected to control by the aforementioned processing system **1312**. As evident from FIGS. **3**, **4**, and **5**, operator display and control **1445**, **1460** may be safely located outside of the under-fill machine **302**, such as at the front portion thereof, and in communication with the processing system **1312**. Further, an input/output “cabinet” **330** may be provided for user access safely at the rear of the machine **302** as shown, such as to the processing system **1312** and/or electromechanical aspects of the machine **302**. This display and control **1445**, **1460** may additionally allow for oversight of the calibration aspects discussed throughout, or may be communicative with one or more secondary stations at which the calibrations discussed throughout are carried out.

(26) With particular reference to FIG. **5**, the referenced right and left robots **304a**, **304b** are shown at the front aspect of the illustrated under-fill machine **302**. Also shown is a circuit board **310** placed upon a carrier **316** at the front of the machine **302** in preparation for insertion into the accommodating input **314**. Of note, the carrier **316** shown, when placed in communication with certain electronic elements, may cause the carrier **316** to serve as the lower heater **322** discussed throughout.

(27) Also shown in FIG. 5 are left and right dispense controllers **502a**, **502b** that allow, such as via one or more communicative connections, for execution of the dispensing of under-fill by the dispensing heads **504a**, **504b** associated, such as by one or more electromechanical connections, with each of the right and left robots **304a**, **304b**. Upon dispensing of the under-fill to components on the circuit board **310**, the circuit board components may be subjected to capillary action of the under-fill due to the heating of at least the lower heater **322** discussed throughout, and, in some embodiments, indexing of the board **310** to subject the ones of the components to the overhead heater **320**. Further, in certain embodiments, components on certain aspects of the board **310** may be subjected to the overhead heater **320** for preheating, and then indexed out of the overhead heater **320** to be subjected to the dispensing heads **504a**, **504b**. And yet further, in some embodiments, underfill may be provided, and capillary action caused by the lower heater **322**, and thereafter aspects of the board **310** may be indexed into association with the overhead heater **320** for purposes of curing, by way of non-limiting example.

(28) More particularly with respect to FIG. 5, several motor controls may be provided for each robot, such as in conjunction with the dispense controllers **502a**, **502b**. Such motor controls and dispense controls may be actuated in accordance with the predetermined programs/recipes **1490** executed by the processing system **1312**, and/or such actuation may be modified by an operator interacting with the processing system **1312**, such as an operator interfacing with the operator display and control **1445**, **1460** discussed herein.

(29) Of note, the robotics **304a**, **304b** shown may include, attached thereto, one or more end-effectors that are and/or include dispense heads **504a**, **504b** to perform the functions discussed throughout upon the exposed portions, i.e., those portions not underneath the overhead heater **320**, of the printed circuit board **310**. That is, the end effector associated with the robots may be or include any type of dispensing head **504a**, **504b**, by way of non-limiting example, capable of dispensing the underfill in accordance with recipe **1490**, as discussed above. Moreover, both the robot **304** and the dispensing head(s) **504** may be under the control of the disclosed processing system **1312** executing the recipe **1490**.

(30) Further, the left and right robots **304a**, **304b** discussed herein as operating dispense heads **504a**, **504b** have ease of access to operate on the upper facing portion of the circuit board **310** that is indexed into physical proximity with the operative area within the chamber of robots **304a**, **304b**, as will be understood to the skilled artisan in light of the discussion herein. Thereby, indexing portions of the board **310** outside of any overhead heater **320** allows for application of underfill, as discussed throughout, by the robot(s) **304a**, **304b** via dispense heads **504a**, **504b**. And after the application of underfill, indexing/rotation of that radial portion under a secondary heater, such as the overhead heater **320**, may enhance capillary action, and/or may provide curing of board components and/or underfill, by way of non-limiting example.

(31) The staged indexing of the board into a primary or secondary heater, such as a secondary heater in the form of overhead heater **320**, such as for the disclosed underfill process, promotes process stability. For example, the disclosed robotics **304** may operate the dispense heads **504** on only an aspect of the board at a given time, such as working on only a quadrant of a circular board at a given time as discussed above, while the remainder of the board may be cured, preheated, or subjected to temperature maintenance, by the combination of the secondary heater **320** and/or any primary heater **322**.

(32) More particularly and as illustrated in FIG. 6, the dispensing of underfill **702** may be performed by each end effector/dispense head **504** associated with the end of the robot arms **304** distal to dispense controllers **502**. As shown, the underfill end effector/dispense head **504** may be self-contained **704**, such as may include heating and cooling features **706** which help to prevent the underfill **702** to be dispensed from curing or otherwise hardening before it can be suitably dispensed from dispense head **504** onto the circuit board. That is, the heat provided by, by way of non-limiting example, at least one heater, such as the lower and overhead heaters **322**, **320**

discussed herein, to the extent such heat escapes into the chamber of an underfill machine in which dispenser heads **504** reside, may cause a curing/crosslinking of the underfill **702**, in whole or in part, before the underfill **702** is dispensed from the illustrated end effector **504**.

(33) FIGS. **7A** and **7B** illustrate an exemplary dispensing end effector **504** in accordance with some embodiments. As illustrated, the embodiments may include a heat shield **802** that may, in part, preclude the underfill **702** as-yet undispensed in the dispense head **504** from undesired curing, i.e., crosslinking, due to a heightened temperature within the chamber. Further illustrated in FIG. **7** is an air purge **804** that may be or form part of a cooling and depressurization system for the enclosed dispense head **504**. The cooling system may include, by way of example, passive or forced recirculation of air, nitrogen, or the like, using, in part, air purge **804**, by way of non-limiting example. Heat shielding **802** and cooling **804** of the dispenser head **504** may prevent the need for variations in underfill processes, such as variations in dispense rates or the useful life of undispensed underfill, which may arise due to an undesirable expedited curing of the underfill while still undispensed from dispense head **504**.

(34) Further illustrated in FIG. **7** may be one or more cameras **810** within the end effector enclosure **704**. These cameras **810** may allow for calibration, as further discussed herein, underfill dispense process and dispense head **504** operation monitoring, and so on.

(35) FIG. **7** additionally illustrate a laser distance sensor **820**, such as may be employed to calibrate the dispense head **504** spatial orientation and/or to periodically, semi-continuously, or continuously monitor the distance from the dispense head **504** to the subject printed circuit board on which the dispensing occurs. Monitoring of the distance from the dispense head **504** to the circuit board allows for enhanced process control, at least in that, to the extent undesirable expedited curing is prevented, underfill dispense rates and locations can be more carefully controlled than in the known art.

(36) FIG. **7** further illustrate lighting **830**, such as may be used for additional process monitoring. Also shown is one or more additional heat barriers **834** to prevent the causation of undesirable expedited curing by heat contamination with enclosure **704**, and it should be noted that additional heat barriers **834** may be opaque or transparent, such as wherein the one or more additional heat barriers **834** are placed within the path of the laser distance sensor **820**, the lighting **830**, or the like.

(37) Yet further, it will be apparent that the dispensing end effector **504** may include one or more different types of dispenser **840**. The dispenser **840** is the ultimate dispensing point from which the underfill **702** discussed herein is dispensed outwardly from the end effector **504** and onto the circuit board. Of note and as shown, the dispenser **840** and/or the end effector **504** may be modular, and as such may include one or more spring or clip actuated releases **850**, whereby actuation of the release **850** may allow for removal and replacement of the dispenser **840**.

(38) By way of non-limiting example, FIGS. **8A**, **8B** and **8C** illustrate particular different dispensers **840** that may be used in the embodiments. FIG. **8A** illustrates a positive pressure needle dispenser **840a**, wherein a needle **902** may be threaded, by way of non-limiting example, onto the dispense head **504** to act as a dispenser **840a**. The needle **902** may be sized so as to provide a flow rate and droplet size unique to a particular embodiment. Similarly, FIG. **8B** illustrates an Archimedes dispenser **840b**, in which actuation of an auger **904** may likewise control underfill flow rate and droplet size.

(39) FIG. **8C** illustrates a piezo jet dispenser **840c**. As illustrated, the piezo jet dispenser **840c** includes a material feed **910**, and piezo-hammer **912** having an angled tip **912a** such that, as the piezo-hammer **912** rises and falls, underfill material **702** from the material feed **910** is forced outward from the dispenser output port **914** onto the circuit board. Thereby, the rise and fall rate of the hammer **912** may be indicative of the dots per second dispensed by the dispenser **840c**, and the angle of the angled tip **912a** of the piezo-hammer **912**, such as in conjunction with the size of the output port **914**, may be indicative of the volume of underfill material dispensed per fall of the piezo-hammer **912**. In accordance with the foregoing, embodiments may comprise a dispensing of

up to 300 drops of underfill per second from the disclosed piezo-jet dispenser **840c**.

(40) Also illustrated in FIG. **8C**, such as completely or partially surrounding the disclosed output port **914**, may be one or more heaters **920**. These heaters **920** may be used to maintain the dispensed underfill material **702** at the proper dispensing temperature for which the underfill material readily flows at the desired flow and droplet rate, but does not reach a temperature at which curing of the underfill material **702** begins prior to dispensing. That is, the one or more heaters **920** at the output port **914** may allow for refined temperature control of the dispensed underfill material **702** with a precision unknown in the prior art.

(41) In accordance with the foregoing, a refined size and number of dots per second of underfill dispensing may be achieved in the embodiments. As will be appreciated, this translates to a particular number of millimeters per second of coverage in the X-Y axes to be provided by each dispensing head **504**. Accordingly, flow rates may be varied as needed, such as for number or size of components on a particular circuit board, or dispensed heads **504** may be modularly employed, such as discussed above, wherein a particular dispense head **504** is used for a particular flow rate of size and number of dots per second. Yet further, each such modular dispense head **504** may have its own unique maximum rate and maximum temperature before clogging occurs, and thus these parameters may be matched to the needs of the processing for a particular circuit board on a case-by-case basis.

(42) FIG. **9** illustrates an exemplary calibration system **950** for a dispense head/end effector **504** and dispenser **840** in accordance with some embodiments. As illustrated, a dispense head **504** may be at the distal end of an underfill dispensing robot arm **304** from a dispense controller **502** that may operate to control the dispense head **504**, as discussed throughout. Further, a calibration system **950** in accordance with the embodiments may include one or more dispense purges **954**, at which air may be expressed from the dispense head **504** so as to avoid clogging and/or undesired curing, and further at which unused or clogged material may be purged from the dispense head **504**.

(43) Further included in the calibration system **950** may be one or more Z-axis height calibration sensors **958**. Such sensors **958** may provide sensing of the height of the dispense head **504** and/or dispense output port **914** from a prospective circuit board, and/or may provide a target location at which the laser height sensor **820** discussed herein as being within the dispense head **504** in some embodiments may assess the propriety of the height sensed above a circuit board.

(44) Further included in an exemplary calibration system **950** may be one or more weighting sensors **970**. At a weighting sensor **970**, each dispensed dot and/or group of dots from dispenser output port **914** may be sensed. Thereby, dots per second and millimeters of coverage per second for a dispense head **540** may be inferred from the sensing by the weighting sensor. Needless to say, such sensing provides for confirmation of a repeatable flow rate and volume for underfill that matches the desired semiconductor process dictated by recipe **1490** for each circuit board.

(45) In conjunction, the aforementioned calibration aspects may allow for variations in the setup of a dispensing head **504**, such as may be entered and/or modified by a user into the user interface and display **1445**, **1460** discussed throughout. As such, dispensing processes of recipe **1490** may be calibrated and recalibrated using the disclosed calibration system **950**. Of course, sensing of proper calibration may also be performed during “real time” execution of underfill, such as by one or more sensors included within the dispense head **504** as discussed herein throughout.

(46) The operation(s) run on the board discussed throughout may comprise a series of process steps encompassed by a software “recipe” **1490** executed by the processing system **1312**. A recipe may be automatically or manually selected, and may execute once the board is associated with the carrier, and/or after the board is entered through the accommodating input. A recipe may consist of a set of defined commands, such as a robot motion, a dispense type, a dispense head temperature, a dispense head distance, a power to a heater, and/or an alignment, by way of example. Commands can be grouped into sub-routines, for example, as will be understood to those skilled in the art.

(47) By way of example, a recipe **1490** may include loading of the board into the accommodating

input; bringing the lower heater to a certain temperature once the board is within the chamber, or prior to the board entering the chamber; moving the robots to each position defined in the recipe to dispense underfill based on controlled operation of dispense heads **504**; and actuating the chamber heater(s) **320, 322** to a predetermined temperature for any aspect(s) of the board indexed associated therewith. In some embodiments, distinct sub-recipes may be run by each of multiple robots **304** and/or dispense heads **504**, in series or in parallel, all running as aspects of the processing system. That is, a series of recipes can be concatenated together into a single recipe, and/or recipes may be parallel or sequenced for proper operation, such as by the processing system and/or a “teaching” mode.

(48) A series of movements, dispense, dispense start and stop, speed and dwell teachings may comprise a discontinuous, semi-continuous, or continuous “path” executed pursuant to a recipe **1490**. In embodiments, the paths of multiple robots may necessarily be deconflicted as part of recipe **1490**, such as to perform collision avoidance between multiple robots having the capabilities to simultaneously take the same position in three-dimensional space, and/or so as to avoid a “double dispense” of underfill by multiple heads **504**.

(49) FIG. **10** depicts an exemplary computer processing system **1312** for use in association with the embodiments, by way of non-limiting example. Processing system **1312** is capable of executing software, such as an operating system (OS), training applications, user interface, and/or one or more other computing algorithms/applications **1490**, such as the recipes discussed herein. The operation of exemplary processing system **1312** is controlled primarily by these computer readable instructions/code **1490**, such as instructions stored in a computer readable storage medium, such as hard disk drive (HDD) **1415**, optical disk (not shown) such as a CD or DVD, solid state drive (not shown) such as a USB “thumb drive,” or the like. Such instructions may be executed within central processing unit (CPU) **1410** to cause system **1312** to perform the disclosed operations, comparisons and calculations. In many known computer servers, workstations, personal computers, and the like, CPU **1410** is implemented in an integrated circuit called a processor.

(50) It is appreciated that, although exemplary processing system **1312** is shown to comprise a single CPU **1410**, such description is merely illustrative, as processing system **1312** may comprise a plurality of CPUs **1410**. Additionally, system **1312** may exploit the resources of remote CPUs (not shown) through communications network **1470** or some other data communications means **1480**, as discussed throughout.

(51) In operation, CPU **1410** fetches, decodes, and executes instructions from a computer readable storage medium, such as HDD **1415**. Such instructions may be included in software **1490**. Information, such as computer instructions and other computer readable data, is transferred between components of system **1312** via the system's main data-transfer path. The main data-transfer path may use a system bus architecture **1405**, although other computer architectures (not shown) can be used.

(52) Memory devices coupled to system bus **1405** may include random access memory (RAM) **1425** and/or read only memory (ROM) **1430**, by way of example. Such memories include circuitry that allows information to be stored and retrieved. ROMs **1430** generally contain stored data that cannot be modified. Data stored in RAM **1425** can be read or changed by CPU **1410** or other hardware devices. Access to RAM **1425** and/or ROM **1430** may be controlled by memory controller **1420**.

(53) In addition, processing system **1312** may contain peripheral communications controller and bus **1435**, which is responsible for communicating instructions from CPU **1410** to, and/or receiving data from, peripherals, such as peripherals **1440, 1445**, and **1450**, which may include printers, keyboards, and/or the operator interaction elements discussed herein throughout. An example of a peripheral bus is the Peripheral Component Interconnect (PCI) bus that is well known in the pertinent art.

(54) Operator display **1460**, which is controlled by display controller **1455**, may be used to display

visual output and/or presentation data generated by or at the request of processing system **1312**, such as responsive to operation of the aforementioned computing programs/applications **1490**. Such visual output may include text, graphics, animated graphics, and/or video, for example. Display **1460** may be implemented with a CRT-based video display, an LCD or LED-based display, a gas plasma-based flat-panel display, a touch-panel display, or the like. Display controller **1455** includes electronic components required to generate a video signal that is sent to display **1460**. (55) Further, processing system **1312** may contain network adapter **1465** which may be used to couple to external communication network **1470**, which may include or provide access to the Internet, an intranet, an extranet, or the like. Communications network **1470** may provide access for processing system **1312** with means of communicating and transferring software and information electronically. Additionally, communications network **1470** may provide for distributed processing, which involves several computers and the sharing of workloads or cooperative efforts in performing a task, as discussed above. Network adaptor **1465** may communicate to and from network **1470** using any available wired or wireless technologies. Such technologies may include, by way of non-limiting example, cellular, Wi-Fi, Bluetooth, infrared, or the like. (56) In the foregoing Detailed Description, it can be seen that various features are grouped together in a single embodiment for the purpose of clarity and brevity of the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the embodiments require more features than are expressly recited herein. Rather, the disclosure is to encompass all variations and modifications to the disclosed embodiments that would be understood to the skilled artisan in light of the disclosure.

Claims

1. A dispensing end effector suitable for dispensing underfill to components on a printed circuit board within an underfill chamber, comprising: an electromechanical connection to at least two dispensing robot arms capable of physically situating the dispensing proximate to the circuit board; a communicative connection to a dispense controller capable of communicatively controlling at least the dispensing; a dispenser which includes an underfill output port, which is capable of the dispensing, and which is removably mounted to the electromechanical connection, wherein the dispenser comprises a piezo-jet dispenser including a piezo-hammer, wherein the piezo-hammer comprises a cantilevered arm in a horizontal orientation and an angled tip positioned on an underside of the arm proximate to a first end; wherein the dispenser comprises a material feed proximate to the tip of the piezo-hammer, configured to feed underfill material between the tip of the piezo-hammer and the output port, wherein at least a portion of the material feed is below the arm of the piezo-hammer when the piezo-hammer is in a risen position; and a protective enclosure at least substantially about the dispenser, including at least one calibration camera and at least one monitoring camera.
2. The dispensing end effector of claim 1, wherein the piezo-jet dispenser dispenses the underfill at a rate of between about 100 and about 300 drops per second.
3. The dispensing end effector of claim 1, wherein the angle of the angled tip of the piezo-hammer of the piezo-jet dispenser at least partially dictates a flow rate of the underfill.
4. The dispensing end effector of claim 1, wherein the protective enclosure is substantially sealed.
5. The dispensing end effector of claim 1, wherein the protective enclosure comprises at least heat shielding.
6. The dispensing end effector of claim 5, wherein the heat shielding is opaque.
7. The dispensing end effector of claim 5, wherein the heat shielding is transparent to allow for process monitoring.
8. The dispensing end effector of claim 1, wherein the protective enclosure comprises at least cooling.

9. The dispensing end effector of claim 8, wherein the cooling comprises an air purge.
 10. The dispensing end effector of claim 9, wherein the air purge comprises a depressurization for the protective enclosure.
 11. The dispensing end effector of claim 8, wherein the cooling comprises one of a forced circulation of air or nitrogen.
 12. The dispensing end effector of claim 1, wherein the protective enclosure comprises a laser distance sensor suitable to spatially orient the dispenser.
 13. The dispensing end effector of claim 12, wherein the spatial orientation comprises a distance from the dispenser to the circuit board.
 14. The dispensing end effector of claim 12, wherein the laser distance sensor is configured to spatially orient the dispenser through a transparent heat barrier.
 15. The dispensing end effector of claim 1, wherein the protective enclosure comprises lighting suitable for process monitoring.
 16. The dispensing end effector of claim 15, wherein the lighting is configured to illuminate through a transparent heat barrier.
 17. The dispensing end effector of claim 1, wherein the dispenser is modular.
 18. The dispensing end effector of claim 1, wherein the electromechanical connection is modular.
 19. The dispensing end effector of claim 1, wherein the output port comprises a heater.
 20. The dispensing end effector of claim 19, wherein the heater is at least substantially about an opening of the output port.
 21. The dispensing end effector of claim 1, further comprising a processing system that indexes the dispensing.
 22. The dispensing end effector of claim 21, wherein the indexing is for a quadrant of the printed circuit board.
 23. The dispensing end effector of claim 1, wherein the dispenser includes one or more spring or clip actuated releases, configured to allow for removal and replacement of the dispenser.
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